

RISK MANAGEMENT

Project # 2

Scenario

GWP2 is a continuation of the work you submitted for GWP1. That is, you are going to work towards the second and third sections of your mini-capstone. Recall that each student individually reads the paper:

Alvi, Danish A. *Application of Probabilistic Graphical Models in Forecasting Crude Oil Price*. 2018. University College London, Dissertation.

<https://arxiv.org/abs/1804.10869>

By the end of GWP2, you will have been deeply immersed in the Bayesian network methodology and how the model gets developed step by step. When completed, GWP2 looks like a literature review, as well as the methodology section of a mini-capstone project.

When all GWPs are completed, the combined three projects look like a mini-capstone.

Overview of GWPs

GWP 1: Problem Formulation. Data Collection.

GWP 2: Methodology Description. Model Development.

GWP 3: Interpretation of Results. Improving the Model.

Tasks

You will be implementing the four stages of Hidden Markov Model Algorithms.

Step 1

- a. Student A writes pseudocode for the forward algorithm and the backward algorithm, as well as a toy example.
 - b. Student B writes pseudocode for the backward Viterbi algorithm, as well as a toy example.
 - c. Student C writes pseudocode for the Baum-Welch algorithm, as well as a toy example
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Step 2

- a. Student A will identify and illustrate examples of bull regimes.
- b. Student B will identify and illustrate examples of bear regimes.
- c. Student C will identify and illustrate examples of stagnant regimes.

Step 3

As a group, the team writes their own version of the paper's Definition 3: Hidden Markov Model (see section 2.1.2.1 in the dissertation). The team will find a different/updated reference. This should be no more than 1–2 pages.

Step 4

For the design, each person carefully studies Figure 3.1 from the paper.

- a. Student A writes up to 5 pages that discuss the macro research, indicator identification, dataset retrieval, and data cleaning process.
- b. Student B writes up to 5 pages that discuss the regime process and explain how the relevant code from *hmms* works.
- c. Student C writes up to 5 pages that discuss how the network is trained using *pgmpy* and how the parameters are tested and validated.

Step 5

As a group, the three papers are combined to form one narrative that should be between 10–15 pages.

Step 6

For regime detection, each person carefully re-reads 4.2 (through 4.2.5) of the dissertation.

- a. Student A writes 1–2 pages explaining the process of transforming the time series.
 - b. Student B writes 1–2 pages explaining how the parameters are learned. The Baum-Welch algorithm can be treated as a black box.
 - c. Student C writes 1–2 pages on finding the most likely sequence of hidden states as the underlying regimes that generated the sequence of observed emissions. The Viterbi algorithm should be treated as a black box.
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Step 7

As a group, the team identifies the latent meaning behind each hidden state.

Step 8

For the Hill working example, each person carefully re-reads Section 4.2.6 of the paper.

Step 9

As a team, the group:

- a. implements the minimal working example of a Hill Climb search, and
 - b. identifies the latent meaning behind each hidden state.
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